

Database Schema — Data Dictionary

Complete reference for the COMPASS PostgreSQL database: **15 application tables** (plus `alembic_version`) in the `public` schema. Every table and every column is documented below. Canonical definitions live in `app/Backend/models.py`.

Native deployment: port `:5433`, database `prostatecancer_db_native`. Docker mode: `:5432`.

How to read the column tables: `Key` column — PK primary key, `FK→t` foreign key to table `t`, `U` part of a unique constraint. `Null` — whether the column may be empty.

The parent-child (anchor) pattern

Two tables hold **identity only** and act as anchors that other tables reference by foreign key: - `transcript_analysis_log` (PK `id`) — one row per pipeline run; every NLP/AI result row points back to it via `analysis_id`. Deleting a run cascade-deletes all its results. - `patient_summary` (PK `file`, `speaker`) — one row per patient; the patient's survey submissions reference it. This gives referential integrity, cascade delete, and survival across re-processing.

Group 1 — Pipeline persistence (7 tables)

The NLP + AI pipeline output. All are tied to one run via `analysis_id → transcript_analysis_log.id`.

transcript_analysis_log — pipeline run header · anchor

One row per analysis run of one transcript file. Parent of every pipeline result table.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment run id. Used as <code>analysis_id</code> by all child tables.
patient_id	varchar(255)	no		Patient identifier (hashed) parsed from the file name.
total_sentences	integer	no		Number of sentences the transcript segmented into.
top_n	integer	no		How many top sentences per model were selected (e.g. 10).
context_window	integer	no		How many neighboring sentences of context were attached (default 3).
model_results	jsonb	yes		Per-model NLP result payload (validated JSON) kept for reference.
xlsx_data	bytea	yes		The exported results xlsx stored inline, so downloads work without a filesystem.
source_filename	varchar(500)	yes		Original uploaded file name.
pipeline_started_at	timestamptz	yes		When the pipeline orchestrator began this file.
analyzed_at	timestamptz	yes		When NLP results were saved (default now()).
ai_overall_score	double	yes		Mean AI score across domains, written when the AI stage finishes.
processed	boolean	yes		True once the full AI pipeline completed for this run.
processed_at	timestamptz	yes		When the AI stage completed.
doctor_id	varchar(255)	yes		Doctor identifier (hashed) parsed from the file name.

nlp_all_predictions — every sentence × 5 model scores

One row per sentence, carrying all five model probabilities. The full raw NLP output.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
analysis_id	integer	no	FK→transcript_analysis_log	The run this sentence belongs to.
patient_id	varchar(255)	no		Patient identifier (denormalized for fast filtering).
sentence_index	integer	no		Global sentence position in the transcript.
utterance_index	integer	no		Utterance (turn) number the sentence came from.
sentence_in_utterance	integer	no		Position of the sentence within its utterance.
speaker	varchar(255)	yes		Speaker label (doctor/patient) of the sentence.
sentence_text	text	yes		The sentence text.
pred_cp	double	yes		Probability the sentence is about Cancer Prognosis.
pred_le	double	yes		Probability — Life Expectancy.
pred_ed	double	yes		Probability — Erectile Dysfunction.
pred_inc	double	yes		Probability — Urinary Incontinence.
pred_ius	double	yes		Probability — Irritative Urinary Symptoms.
created_at	timestamptz	no		Row insert time (default now()).

(pred_* are NLP probabilities in [0,1], not scores.)

sentence_prediction — top-N selected sentences (per model)

The per-domain representative sentences shown to the doctor. One row per (sentence × model) that made a model's top-N.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
analysis_id	integer	no	FK→transcript_analysis_log	The run.
patient_id	varchar(255)	no		Patient identifier (denormalized).
model	varchar(10)	no		Domain/model this row is for: <code>cp / le / ed / inc / ius</code> .
sentence_index	integer	no		Global sentence position.

Column	Type	Null	Key	Description
utterance_index	integer	no		Utterance number.
sentence_in_utterance	integer	no		Position within the utterance.
speaker	varchar(100)	yes		Speaker label.
sentence_text	text	yes		The selected sentence text.
pred_score	double	no		This model's probability for the sentence (the ranking value).
context	text	yes		Neighboring sentences; the focus sentence wrapped in <code><main>_</main></code> .

nlp_pipeline_intermediate — NLP step-state snapshots

One JSONB snapshot per NLP step for debugging / reproduction.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
analysis_id	integer	no	FK→transcript_analysis_log	The run.
patient_id	varchar(255)	no		Patient identifier.
step	varchar(20)	no		NLP step name: <code>raw / filtered / sentences / top_by_model</code> (CHECK-constrained).
payload	jsonb	no		The step's input/output data.
row_count	integer	yes		Number of rows in the payload (quick size indicator).
created_at	timestamptz	no		Insert time.

llm_pipeline_intermediate — per-domain AI intermediate state

One row per AI candidate (domain × sentence) recording how the AI scored/filtered it.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
analysis_id	integer	no	FK→transcript_analysis_log	The run.
patient_id	varchar(255)	no		Patient identifier.
domain	varchar(10)	no		Domain: <code>cp / le / ed / inc / ius</code> (CHECK).
step	varchar(20)	no		AI step; currently always <code>extraction</code> (CHECK).
sentence_index	integer	no		Global sentence position of the candidate.
sentence_text	text	yes		Candidate sentence text.
context	text	yes		Surrounding context.
pred_score	double	yes		Upstream NLP probability of the candidate.
ai_score	smallint	yes		GPT-4o score 0-5 for the candidate.
score_explanation	text	yes		The model's reasoning for the score.
estimate	text	yes		Extracted numeric estimate (long text; TEXT to avoid overflow).
treatment	text	yes		Extracted treatment context.
survived_filter	boolean	no		Whether this candidate passed the filter step (default false).
created_at	timestamptz	no		Insert time.

llm_domain_scoring_and_summary — final patient-visible AI output

One row per domain (a domain may have several rows when the treatment branches). This is what the patient/doctor screens display.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
analysis_id	integer	no	FK→transcript_analysis_log	The run.
patient_id	varchar(255)	no		Patient identifier.
domain	varchar(10)	no		Domain: <code>cp / le / ed / inc / ius</code> .
ai_score	integer	yes		Final 0-5 AI score for the domain.
score_explanation	text	yes		Reasoning behind the score.
extracted_estimate	text	yes		The final extracted numeric estimate.
treatment	text	yes		Treatment this row pertains to (for treatment-branched domains).
source_sentence	text	yes		The consultation sentence the AI chose as the source.
source_context	text	yes		Context with <code><main></code> markers; the screen's "From your consultation".
reformat_sentence	text	yes		The patient-friendly rewrite shown in the report.
source_filename	varchar(500)	yes		Original file name (denormalized).
created_at	timestamptz	yes		Insert time.

patient_summary — patient parent key · anchor

One row per patient. Holds only the identity; patient survey submissions FK to it.

Column	Type	Null	Key	Description
file	varchar(255)	no	PK	The patient's file name (patient+doctor+date key).

Column	Type	Null	Key	Description
speaker	varchar(100)	no	PK	The patient speaker label inside that file.

Group 2 — Behavior tracking (3 tables)

UI-interaction events (how the user navigated), not answer values. All three share `session_id`, `file`, `speaker`, `event_type`, `metadata`, `device_type`, `client_timestamp`, `created_at` plus per-area extras. `event_type` is CHECK-constrained to each area's vocabulary.

patient_report_page_behavior — first-visit report page behavior

Events on the first-visit **report** page (reading AI summaries). Report-only.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
session_id	varchar(100)	no		One id per page mount; groups a visit's events.
file	varchar(255)	no		Patient file.
speaker	varchar(100)	no		Patient speaker.
event_type	varchar(30)	no		Interaction type: <code>page_view</code> / <code>topic_open</code> / <code>topic_close</code> / <code>evidence_open</code> / <code>evidence_close</code> / <code>summary_open</code> / <code>summary_close</code> / <code>rating_click</code> / <code>slider_moved</code> / <code>answer_changed</code> / <code>domain_submit</code> (CHECK).
domain	varchar(50)	yes		Domain the event is about (<code>cp</code> / <code>le</code> / <code>ed</code> / <code>inc</code> / <code>ius</code>); required for topic/evidence/rating events.
rating	smallint	yes		1-5 rating (for <code>rating_click</code> ; CHECK 1-5).
metadata	jsonb	no		Free-form event details (timing, screen, etc.); default <code>{}</code> .
device_type	varchar(20)	yes		<code>mobile</code> / <code>tablet</code> / <code>desktop</code> .
client_timestamp	timestampz	no		When the event happened on the client.
created_at	timestampz	no		Server insert time.

patient_followup_survey_page_behavior — survey page behavior

Events on the survey pages: follow-up (SDM/DCS/Risk/Satisfaction) **and** the first-visit Risk survey.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
session_id	varchar(100)	no		Session grouping id.
file	varchar(255)	no		Patient file.
speaker	varchar(100)	no		Patient speaker.
event_type	varchar(30)	no		<code>page_view</code> / <code>survey_step_view</code> / <code>survey_answer</code> / <code>survey_complete</code> / <code>session_end</code> (+ the report-page vocabulary for the embedded Risk step); CHECK.
survey_type	varchar(30)	yes		Which survey: <code>sdm</code> / <code>dcs</code> / <code>risk_perception</code> / <code>satisfaction</code> (CHECK). Required for <code>survey_answer</code> .
question_id	varchar(50)	yes		Question the event refers to. Required for <code>survey_answer</code> .
step_number	smallint	yes		Survey step index (only for <code>survey_step_view</code>).
metadata	jsonb	no		Free-form details; default <code>{}</code> .
device_type	varchar(20)	yes		Device type.
client_timestamp	timestampz	no		Client event time.
created_at	timestampz	no		Server insert time.
domain	varchar(50)	yes		Domain (used by the embedded first-visit Risk step).
rating	smallint	yes		Rating (used by the embedded Risk step).

doctor_behavior — doctor dashboard behavior

Events on the doctor consultation dashboard.

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
session_id	varchar(100)	no		Session grouping id.
file	varchar(255)	yes		Patient file (null on the dashboard list view).
speaker	varchar(100)	no		Doctor identifier.
event_type	varchar(30)	no		<code>page_view</code> / <code>view_change</code> / <code>patient_select</code> / <code>topic_select</code> / <code>sentence_select</code> / <code>rewrite_open</code> / <code>rewrite_input</code> / <code>rewrite_apply</code> / <code>rubric_open</code> / <code>rubric_close</code> / <code>rubric_score</code> (CHECK).
target_type	varchar(20)	yes		What the event targets: <code>patient</code> / <code>topic</code> / <code>sentence</code> (CHECK).
target_id	varchar(255)	yes		Identifier of the target (patient/topic/sentence).
metadata	jsonb	no		Free-form details; default <code>{}</code> .
device_type	varchar(20)	yes		Device type.
client_timestamp	timestampz	no		Client event time.
created_at	timestampz	no		Server insert time.

Group 3 — Authentication (2 tables)

auth_user — accounts

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment user id.
username	varchar(150)	no		Login name.
email	varchar(255)	yes	U	Email (unique).
password_hash	varchar(255)	yes		Hashed password (null for external auth providers).

Column	Type	Null	Key	Description
role	varchar(20)	no		admin / user / readonly (CHECK); default user.
is_superuser	boolean	no		Superuser flag; default false.
is_active	boolean	no		Whether the account is enabled; default true.
auth_provider	varchar(50)	no		local or an external provider; default local.
created_at	timestamptz	yes		Creation time.
updated_at	timestamptz	yes		Last update time.

auth_api_key — API keys

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment key id.
user_id	integer	no	FK→auth_user	Owner (cascade delete).
key_hash	varchar(255)	no		Hashed API key (raw key never stored); compared with hmac.compare_digest.
label	varchar(100)	yes		Human label for the key.
is_active	boolean	no		Whether the key is enabled; default true.
created_at	timestamptz	yes		Creation time.
expires_at	timestamptz	yes		Optional expiry.
last_used_at	timestamptz	yes		Last time the key authenticated.

Group 4 — Other (3 tables)

patient_survey_submission_log — all patient survey answers

One row per submission. The single source of truth for every patient survey answer: follow-up surveys (sdm / dcs / satisfaction) and the first-visit Risk cognition survey (risk_perception_2). FK to patient_summary .

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
file	varchar(255)	no	FK→patient_summary	Patient file.
speaker	varchar(100)	no	FK→patient_summary	Patient speaker.
survey_type	varchar(50)	no		risk_perception_2 (first-visit Risk), sdm , dcs , satisfaction , ...
answers	jsonb	no		The submitted answers (shape depends on survey_type: risk_perception_2 nests domain→question_id).
extra_data	jsonb	yes		Optional extra payload.
submitted_at	timestamptz	yes		Submission time (default now()).
redcap_synced	boolean	yes		Whether the row was pushed to REDCap; default false.
redcap_record_id	varchar(255)	yes		REDCap-side record id after sync.
redcap_error	text	yes		Last REDCap sync error (for retry), if any.

session_recording — consultation audio/replay chunks

Column	Type	Null	Key	Description
id	integer	no	PK	Auto-increment row id.
session_id	varchar(100)	no		Recording session id.
chunk_index	integer	no		Ordering index of this chunk; default 0.
file	varchar(255)	yes		Patient file the recording belongs to.
visit_type	varchar(20)	yes		Which visit produced it.
recording_data	bytea	yes		The raw recorded bytes.
event_count	integer	no		Number of captured events in the chunk; default 0.
created_at	timestamptz	yes		Insert time.
area	varchar(40)	no		Capture area: patient_first / patient_first_report / patient_first_survey / patient_followup / doctor / physician / unknown (CHECK).

doctor_rewrite_log — AI-assisted sentence rewrites

One row per rewrite. Composite PK (file, i, i2, time) keeps every revision as a separate audit row.

Column	Type	Null	Key	Description
file	varchar(255)	no	PK	Patient file.
i	integer	no	PK	Utterance index of the sentence.
i2	integer	no	PK	In-utterance index of the sentence.
time	timestamptz	no	PK	Rewrite time — part of the PK so revisions don't overwrite each other.
speaker	varchar(100)	yes		Speaker label of the sentence.
original_sentence	text	yes		The sentence before the rewrite.
revised_sentence	text	yes		The doctor's rewritten sentence.
score	double	yes		Score associated with the rewrite.
class	varchar(100)	yes		Domain/class of the sentence.

Group 5 — System (1 table)

Column	Type	Null	Key	Description
version_num	varchar(32)	no	PK	The migration revision the DB is currently at. Managed by Alembic, not the app.

Key indexes (beyond primary keys)

- Pipeline: `analysis_id` and `(patient_id, ...)` indexes on the child tables for fast per-run / per-patient reads.
- Behavior: `(session_id)`, `(file, event/survey)`, `(client_timestamp)` on each behavior table.
- Surveys: `(file, submitted_at)`, `(survey_type)`, `(speaker)` on `patient_survey_submission_log`.
- Auth: `(user_id)`, `(key_hash)` on api keys; `(username)` on users.

See Also

- [ARCHITECTURE.md](#) — system overview · [DB_TABLES_ROLES.md](#) — the story of one file's journey · [AI_PIPELINE_NLP_DB_TABLES.md](#) — pipeline→table mapping · [INDEX.md](#) — one-page overview.
- [app/Backend/models.py](#) — canonical SQLAlchemy definitions.